# Jongseo Lee

Vision and Learning Lab, Kyung Hee University, South Korea

jong980812@khu.ac.kr | jong980812.github.io | vll.khu.ac.kr | jong980812 | jongseo-lee-a551ab244

## Research Interests

- **Trustworthy video understanding:** interpretable concept-based models; diagnosing and improving Video-LLMs (motion and physical reasoning)
- **Video world models for Physical AI:** learning to perceive and predict physical dynamics from video (intuitive physics)
- **Video representation learning:** multimodal (audio-visual, video-language) and continual learning

## Education

**Researcher (Post-M.S.)**, Vision and Learning Lab, Kyung Hee University, South Korea — Sep. 2025 – Present
Full-time research position | Advisor: Prof. Jinwoo Choi

**M.S. in Computer Science and Engineering**, Kyung Hee University, South Korea — Sep. 2023 – Aug. 2025
Advisor: Prof. Jinwoo Choi (Vision and Learning Lab)
Degree: Master of Engineering | GPA: 4.21 / 4.3

**B.S. in Biomedical Engineering & Electronic Engineering**, Kyung Hee University, South Korea — Mar. 2017 – Aug. 2023
Double major | Degree: Bachelor of Engineering | GPA: 3.95 / 4.3

## Publications

(* equal contribution, † corresponding author)

**International Conferences & Journals**

1. **Which Way Did It Move? Diagnosing and Overcoming Directional Motion Blindness in Video-LLMs**
   **Jongseo Lee***, Hyuntak Lee*, Sunghun Kim, Sooa Kim, Jihoon Chung, Jinwoo Choi†
   **NeurIPS 2026** [paper] [project] [code]
   Also presented at the 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents (CompLearn), ICML 2026 (non-archival poster)

2. **TRACE: Trajectory-Aware Concepts for Explainable Video Understanding**
   Wooil Lee*, Junpyo Hong*, **Jongseo Lee**, Tuomas Oikarinen, Tsui-Wei Weng, Jinwoo Choi†
   **NeurIPS 2026**

3. **Disentangled Concepts Speak Louder Than Words: Explainable Video Action Recognition**
   **Jongseo Lee**, Wooil Lee, Gyeong-Moon Park†, Seong Tae Kim†, Jinwoo Choi†
   **NeurIPS 2025, Spotlight** (top 3.5%) [paper] [code]

4. **ESSENTIAL: Episodic and Semantic Memory Integration for Video Class-Incremental Learning**
   **Jongseo Lee***, Kyungho Bae*, Kyle Min, Gyeong-Moon Park†, Jinwoo Choi†
   **ICCV 2025, Highlight Presentation** (top 2.9%) [paper] [code]

5. **CA$^2$ST: Cross-Attention in Audio, Space, and Time for Holistic Video Recognition**
   **Jongseo Lee***, Joohyun Chang*, Dongho Lee, Jinwoo Choi†
   **IEEE TPAMI, 2026** [paper]

6. **PCBEAR: Pose Concept Bottleneck for Explainable Action Recognition**
   **Jongseo Lee**, Wooil Lee, Gyeong-Moon Park†, Seong Tae Kim†, Jinwoo Choi†
   **CVPR Workshop on XAI4CV 2025, Spotlight** (top 16.7%) [paper]

7. **CAST: Cross-Attention in Space and Time for Video Action Recognition**
   **Jongseo Lee***, Dongho Lee*, Jinwoo Choi†
   **NeurIPS 2023** [paper] [poster] [code]

**Preprints**

8. **PCEvE: Part Contribution Evaluation Based Model Explanation for Human Figure Drawing Assessment and Beyond**
   **Jongseo Lee***, Geo Ahn*, Seong Tae Kim†, Jinwoo Choi†
   arXiv preprint, 2024 [paper]

**Non-Archival Workshop Presentations**

- **BiAdA: Holistic Video Recognition via Bidirectional Adapter Fusion**
  **Jongseo Lee***, Joohyun Chang*, Dongho Lee, Jinwoo Choi†
  ICCV 2025 Workshop on Long Video Foundations (LongVidFoundations), non-archival poster

**Domestic Publications**

- **Metaverse Interface with Haptic and Rigid Sense Feedback at a Low Cost**
  **Jongseo Lee**, Su Hyeon Kim, Sun Woong Jang, Jun Yeong Moon, Doug Young Suh†
  Journal of Appropriate Technology, 8(2), 2022 [paper]

## Research Experience

**Researcher (Post-M.S.)**, Vision and Learning Lab, Kyung Hee University, South Korea — Sep. 2025 – Present
Leading research on video-LLM evaluation/diagnosis and explainable video understanding; international collaborations with UC San Diego (Prof. Tsui-Wei Weng, TRACE) and Princeton University (Jihoon Chung, DeltaDirect).

**Graduate Research Assistant**, Vision and Learning Lab, Kyung Hee University, South Korea — Sep. 2023 – Aug. 2025
Researched video class-incremental learning and explainable action recognition; collaborated with Intel Labs (Kyle Min) on ESSENTIAL; worked on ETRI-funded research projects.

**Undergraduate Research Intern**, Vision and Learning Lab, Kyung Hee University, South Korea — Feb. 2022 – Jul. 2023
Conducted research on video action recognition; co-led CAST (NeurIPS 2023) as a co-first author.

## Research Projects

**Skeleton-Based Choreography Recognition for K-POP Dance Videos (SkeletonAR)** — 2026
Industry-funded research project with MetaRocket — skeleton-based recognition of fast, multi-person choreography; KOLAS-accredited third-party performance test in progress

**Motivational Context Analysis of Interaction-Based Reactions in Video** — 2025
Funded research project with ETRI

**Classification and Explanation Models for ASD Screening Based on Drawing Video Understanding** — 2024
Funded research project with ETRI — related publication: PCEvE

**Deep Learning Models and Explainable AI for ASD Screening Drawing Data** — 2023
Funded research project with ETRI

**Rule-Based Group Activity Recognition Video Dataset (AI Hub)** — 2022
National dataset construction project in collaboration with GAZZILABS — developed and validated AI recognition algorithms

## Honors & Awards

- Top Reviewer, NeurIPS 2026 — 2026
- Spotlight Presentation, NeurIPS 2025 (top 3.5% of accepted papers) — 2025
- Highlight Presentation, ICCV 2025 (top 2.9% of accepted papers) — 2025
- Spotlight, CVPR 2025 Workshop on Explainable AI for Computer Vision (XAI4CV) (top 16.7% of accepted papers) 2025

## Scholarships

**Kyung Hee Scholarship for Outstanding Undergraduate Alumni**, Kyung Hee University — Sep. 2023 – Aug. 2025
Full M.S. tuition for 4 semesters, awarded for outstanding undergraduate academic record (approx. $21,600 total)

**Woowa Brothers Scholarship**, JUMP Foundation — 2022
Private foundation scholarship for undergraduate study (approx. $2,900)

## Technical Skills

**Programming:** Python, C/C++, Shell (Bash) **Frameworks:** PyTorch **Tools:** Linux, LaTeX
**Systems:** Large-scale video/multimodal model training and evaluation

## Academic Service

**Reviewer:**

- NeurIPS 2026 (Top Reviewer)
- AAAI 2025
- CVPR Workshop on Explainable AI for Computer Vision (XAI4CV) 2026
- 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents (CompLearn), ICML 2026

## Military Service

**Republic of Korea Army**, Sergeant (active duty) — Jul. 2018 – Mar. 2020
Completed mandatory military service; honorably discharged

## Referees

**Jinwoo Choi**, Assistant Professor, Ph.D.
School of Computing
Kyung Hee University, Yongin, Korea
e-mail: jinwoochoi@khu.ac.kr
*M.S. advisor and current supervisor*

**Seong Tae Kim**, Associate Professor, Ph.D.
School of Computing
Kyung Hee University, Yongin, Korea
e-mail: st.kim@khu.ac.kr
*Collaborator and co-corresponding author*

**Kyle Min**, Principal Applied Scientist, Ph.D.
Oracle
Seattle, WA, USA
e-mail: kylemin@umich.edu
*Research collaborator*

**Gyeong-Moon Park**, Associate Professor, Ph.D.
Department of Artificial Intelligence
Korea University, Seoul, Korea
e-mail: gm-park@korea.ac.kr
*Collaborator and co-corresponding author*

**Kyungho Bae**, Machine Learning Engineer
Karrot (Daangn)
Seoul, Korea
e-mail: backdrop9019@gmail.com
*Co-first author; former lab colleague*

**Tsui-Wei (Lily) Weng**, Assistant Professor, Ph.D.
Halıcıoğlu Data Science Institute
University of California San Diego, La Jolla, CA, USA
e-mail: lweng@ucsd.edu
*Research collaborator*